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UNITED INTERNATIONAL UNIVERSITY EXAM PAPER

INSTITUTION AND COURSE DETAILS

- Institution: United International University
- School: School of Science and Engineering
- Trimester: Summer 2024
- Assessment Type: CT-III
- Course Title: Coordinate Geometry and Vector Analysis
- Course Code: MATH 2201
- Marks: 10
- Total Time: 30 minutes

EXAM QUESTIONS

Answer all questions.

1. Let L1 and L2 be the lines:

Line L1: $x = 2 - t$, $y = 4 + 4t$, $z = 1 + 2t$

Line L2: $x = t$, $y = 3$, $z = 2t$

- i) Are the lines parallel?
- ii) Do the lines intersect?
- iii) Find the distance between the lines.
- iv) Find the intersection point of the line L1 and the plane $x - y + 2z = 0$.

(Total: 10 Marks)